is, the quantities can take on only discrete values. These devices are most often electronic, but they can also be mechanical, magnetic, or pneumatic. Some of the more familiar digital systems include digital computers and calculators, digital audio and video equipment, and the telephone system—the world's largest digital system.

An **analog system** contains devices that manipulate physical quantities that are represented in analog form. In an analog system, the quantities can vary over a continuous range of values. For example, the amplitude of the output signal to the speaker in a radio receiver can have any value between zero and its maximum limit. Other common analog systems are audio amplifiers, magnetic tape recording and playback equipment, and a simple light dimmer switch.

Advantages of Digital Techniques

An increasing majority of applications in electronics, as well as in most other technologies, use digital techniques to perform operations that were once performed using analog methods. The chief reasons for the shift to digital technology are:

1. *Digital systems are generally easier to design.* The circuits used in digital systems are *switching circuits,* where *exact* values of voltage or current are not important, only the range (HIGH or LOW) in which they fall are important.
2. *Information storage is easy.* This is accomplished by special devices and circuits that can latch onto digital information and hold it for as long as necessary, and mass storage techniques that can store billions of bits of information in a relatively small physical space. Analog storage capabilities are, by contrast, extremely limited.
3. *Accuracy and precision are easier to maintain throughout the system.* Once a signal is digitized, the information it contains does not deteriorate as it is processed. In analog systems, the voltage and current signals tend to be distorted by the effects of temperature, humidity, and component tolerance variations in the circuits that process the signal.
4. *Operation can be programmed.* It is fairly easy to design digital systems whose operation is controlled by a set of stored instructions called a *program.* Analog systems can also be *programmed,* but the variety and the complexity of the available operations are severely limited.
5. *Digital circuits are less affected by noise.* Spurious fluctuations in voltage (noise) are not as critical in digital systems because the exact value of a voltage is not important, as long as the noise is not large enough to prevent us from distinguishing a HIGH from a LOW.
6. *More digital circuitry can be fabricated on IC chips.* It is true that analog circuitry has also benefited from the tremendous development of IC technology, but its relative complexity and its use of devices that cannot be economically integrated (high-value capacitors, precision resistors, inductors, transformers) have prevented analog systems from achieving the same high degree of integration.

The real world is analog.
Processing digitized signals takes time.

Most physical quantities are analog in nature, and these quantities are often the inputs and outputs that are being monitored, operated on, and controlled by a system. Some examples are temperature, pressure, position, velocity, liquid level, flow rate, and so on. We are in the habit of expressing these quantities *digitally*, such as when we say that the temperature is 64° (63.8° when we want to be more precise), but we are really making a digital approximation to an inherently analog quantity.

To take advantage of digital techniques when dealing with analog inputs and outputs, three steps must be followed:

1. Convert the real-world analog inputs to digital form.
2. Process *(operate on)* the digital information.
3. Convert the digital outputs back to real-world analog form.

Figure 1-1 shows a block diagram of this for a typical temperature control system. As the diagram shows, the analog temperature is measured and the measured value is then converted to a digital quantity by an **analog-to-digital converter (ADC)**. The digital quantity is then processed by the digital circuitry, which may or may not include a digital computer. Its digital output is converted back to an analog quantity by a **digital-to-analog converter (DAC)**. This analog output is fed to a controller, which takes some kind of action to adjust the temperature.

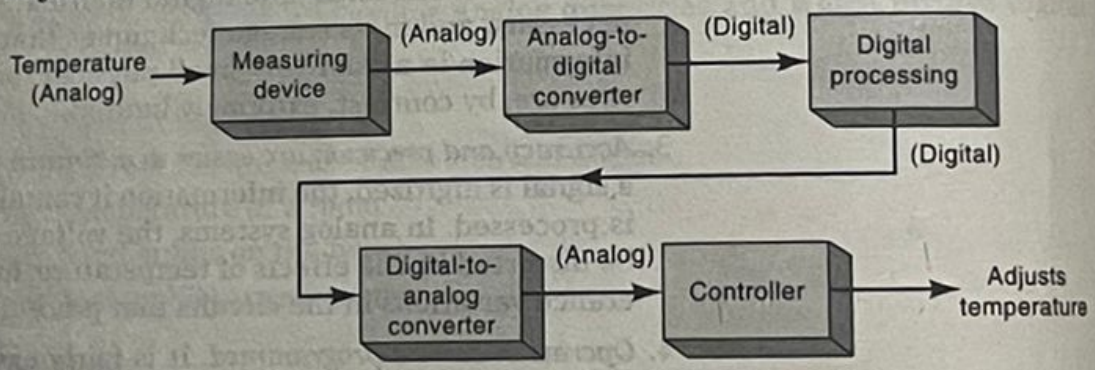


FIGURE 1-1 Block diagram of a temperature control system that requires analog/digital conversions to allow the use of digital processing techniques.

Another good example where conversion between analog and digital takes place is in the recording of audio. Compact disks (CDs) have taken the recording industry by storm because they provide a much better means for recording and playing back music. The process works something like this: (1) sounds from instruments and human voices produce an analog voltage signal in a microphone; (2) this analog signal is converted to a digital format using an analog-to-digital conversion process; (3) the digital information is stored on the CD's surface; (4) during playback, the CD player takes the digital information from the CD surface and converts it into an analog signal that is then

wire that make consistent, intelligent, logical decisions. It is vitally important that we have a method to describe the logical decisions made by these circuits. In other words, we must describe how they operate. In this chapter, we will discover many ways to describe their operation. Each description method is important because all these methods commonly appear in technical literature and system documentation and are used in conjunction with modern design and development tools.

We will begin by studying Boolean algebra, which has been used for hundreds of years as a tool for the analysis and design of logic systems. Boolean algebra is a relatively simple mathematical tool that allows us to describe the relationship between a logic circuit's output(s) and its inputs as an algebraic equation (a Boolean expression). In this chapter, we will study the most basic logic circuits—*logic gates*—which are the fundamental building blocks from which all other logic circuits and digital systems are constructed. We will see how the operation of the different logic gates and the more complex circuits formed from combinations of logic gates can be described and analyzed using Boolean algebra. We will also get a glimpse of how Boolean algebra can be used to simplify a circuit's Boolean expression so that the circuit can be rebuilt using fewer logic gates and/or fewer connections. Much more will be done with circuit simplification in Chapter 4.

Boolean algebra is not only used as a tool for analysis and simplification of logic systems. It can also be used as a tool to create a logic circuit that will produce the desired input/output relationship. This process is often called synthesis of logic circuits as opposed to analysis. Other techniques have been used in the analysis, synthesis, and documentation of logic systems and circuits including truth tables, schematic symbols, timing diagrams, and—last but by no means least—language. To categorize these methods, we could say that Boolean algebra is a mathematic tool, truth tables are data organizational tools, schematic symbols are drawing tools, timing diagrams are graphing tools, and language is the universal description tool.

Today, any of these tools can be used to provide input to computers. The computers can be used to simplify and translate between these various forms of description and ultimately provide an output in the form necessary to implement a digital system. To take advantage of the powerful benefits of computer software, we must first fully understand the acceptable ways for describing these systems in terms the computer can understand. This chapter will lay the groundwork for further study of these vital tools for synthesis and analysis of digital systems.

Clearly the tools described here are invaluable tools in describing, analyzing, designing, and implementing digital circuits. The student who expects to work in the digital field must work hard at understanding and becoming comfortable with Boolean algebra (believe us, it's much, much easier than conventional algebra) and all the other tools. Do *all* of the examples, exercises, and problems, even the ones your instructor doesn't assign. When those run out, make up your own. The time you spend will be well worth it because you will see your skills improve and your confidence grow.

3-1 BOOLEAN CONSTANTS AND VARIABLES

Boolean algebra differs in a major way from ordinary algebra because Boolean constants and variables are allowed to have only two possible values, 0 or 1. A Boolean variable is a quantity that may, at different times, be equal to either 0 or 1. Boolean variables are often used to represent the voltage level present on a wire or at the input/output terminals of a circuit. For example, in a certain digital system, the Boolean value of 0 might be assigned to any voltage in the range from 0 to 0.8 V, while the Boolean value of 1 might be assigned to any voltage in the range 2 to 5 V. Voltages between 0.8 and 2 V are undefined (neither 0 nor 1) and should not occur under normal circumstances.

Thus, Boolean 0 and 1 do not represent actual numbers but instead represent the state of a voltage variable, or what is called its **logic level**. A voltage in a digital circuit is said to be at the logic 0 level or the logic 1 level, depending on its actual numerical value. In digital logic, several other terms are used synonymously with 0 and 1. Some of the more common ones are shown in Table 3-1. We will use the 0/1 and LOW/HIGH designations most of the time.

TABLE 3-1

Logic 0	Logic 1
False	True
Off	On
Low	High
No	Yes
Open switch	Closed switch

As we said in the introduction, **Boolean algebra** is a means for expressing the relationship between a logic circuit's inputs and outputs. The inputs are considered logic variables whose logic levels at any time determine the output levels. In all our work to follow, we shall use letter symbols to represent logic variables. For example, the letter A might represent a certain digital circuit input or output, and at any time we must have either $A = 0$ or $A = 1$; if not one, then the other.

Because only two values are possible, Boolean algebra is relatively easy to work with compared with ordinary algebra. In Boolean algebra, there are no fractions, decimals, negative numbers, square roots, cube roots, logarithms, imaginary numbers, and so on. In fact, in Boolean algebra there are only *three* basic operations: *OR, AND,* and *NOT.*

These basic operations are called *logic operations.* Digital circuits called *logic gates* can be constructed from diodes, transistors, and resistors connected so that the circuit output is the result of a basic logic operation (*OR, AND, NOT*) performed on the inputs. We will be using Boolean algebra first to describe and analyze these basic logic gates, then later to analyze and design combinations of logic gates connected as logic circuits.

3-2 TRUTH TABLES

A **truth table** is a means for describing how a logic circuit's output depends

state. The second entry shows that when input B is changed to the 1 state, so that $A = 0$ and $B = 1$, the output x becomes a 0. In a similar way, the table shows what happens to the output state for any set of input conditions.

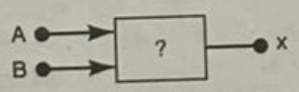
Figures 3-1(b) and (c) show samples of truth tables for three- and four-input logic circuits. Again, each table lists all possible combinations of input logic levels on the left, with the resultant logic level for output x on the right. Of course, the actual values for x will depend on the type of logic circuit.

Note that there are 4 table entries for the two-input truth table, 8 entries for a three-input truth table, and 16 entries for the four-input truth table. The number of input combinations will equal 2^N for an N-input truth table.

FIGURE 3-1 Example truth tables for (a) two-input, (b) three-input, and (c) four-input circuits.

Inputs → A, B; Output → x

A	B	x
0	0	1
0	1	0
1	0	1
1	1	0



(a)

A	B	C	x
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

(b)

A	B	C	D	x
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

(c)

Also note that the list of all possible input combinations follows the binary counting sequence, and so it is an easy matter to write down all of the combinations without missing any.

REVIEW QUESTIONS

1. What is the output state of the four-input circuit represented in Figure 3-1(c) when all inputs except B are 1?
2. Repeat question 1 for the following input conditions: $A = 1, B = 0, C = 1, D = 0$.
3. How many table entries are needed for a five-input circuit?

3-3 OR OPERATION WITH OR GATES

The **OR operation** is the first of the three basic Boolean operations to be learned. The truth table in Figure 3-2(a) shows what happens when two logic inputs, A and B, are combined using the OR operation to produce the output x. The table shows that x is a logic 1 for every combination of input levels where one *or* more inputs are 1. The only case where x is a 0 is when both inputs are 0.

The Boolean expression for the OR operation is

$$x = A + B$$

In this expression, the + sign does not stand for ordinary addition; it stands for the OR operation. The OR operation is similar to ordinary addition except

3-23. Prove theorems (15a) and (15b) by trying all possible cases.

3-24. (a) Simplify the following expression using theorems (13b), (3), and (4):

$$x = (M + N)(\overline{M} + P)(\overline{N} + \overline{P})$$

(b) Simplify the following expression using theorems (13a), (8), and (6):

$$z = \overline{A}B\overline{C} + AB\overline{C} + B\overline{C}D$$

SECTIONS 3-11 AND 3-12

3-25. Prove DeMorgan's theorems by trying all possible cases.

3-26. Simplify each of the following expressions using DeMorgan's theorems.

(a) $\overline{\overline{A}B\overline{C}}$ (d) $\overline{A + \overline{B}}$ (g) $\overline{A(\overline{B + \overline{C}})D}$

(b) $\overline{\overline{A} + \overline{B}C}$ (e) $\overline{\overline{A}\overline{B}}$ (h) $\overline{(M + \overline{N})(\overline{M} + N)}$

(c) $\overline{AB\overline{CD}}$ (f) $\overline{\overline{A} + \overline{C} + \overline{D}}$ (i) $\overline{\overline{A}\overline{B}\overline{C}D}$

3-27. Use DeMorgan's theorems to simplify the expression for the output of Figure 3-46.

3-28. Convert the circuit of Figure 3-44(b) to one using only NAND gates. Then write the output expression for the new circuit, simplify it using DeMorgan's theorems, and compare it with the expression for the original circuit.

3-29. Convert the circuit of Figure 3-44(a) to one using only NOR gates. Then write the expression for the new circuit, simplify it using DeMorgan's theorems, and compare it with the expression for the original circuit.

3-30. Show how a two-input NAND gate can be constructed from two-input NOR gates.

3-31. Show how a two-input NOR gate can be constructed from two-input NAND gates.

3-32. A jet aircraft employs a system for monitoring the rpm, pressure, and temperature values of its engines using sensors that operate as follows:

RPM sensor output = 0 only when speed < 4800 rpm
P sensor output = 0 only when pressure < 220 psi
T sensor output = 0 only when temperature < 200° F

Figure 3-47 shows the logic circuit that controls a cockpit warning light for certain combinations of engine conditions. Assume that a HIGH at output *W* activates the warning light.

(a) Determine what engine conditions will give a warning to the pilot.

(b) Change this circuit to one using all NAND gates.

Table 1.1. Multiples and Sub-multiples of Ohm

Prefix	*Its meaning*	*Abbreviation*	*Equal to*
Mega-	One million	M Ω	10^6 Ω
Kilo-	One thousand	k Ω	10^3 Ω
Centi-	One hundredth	c Ω	10^{-2} Ω
Milli-	One thousandth	m Ω	10^{-3} Ω
Micro-	One millionth	μ Ω	10^{-6} Ω

1.6. Laws of Resistance

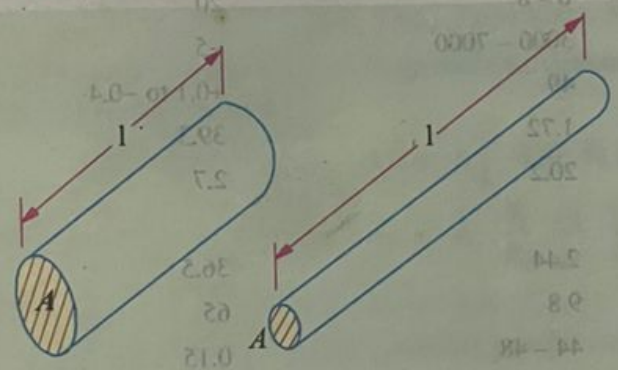
The resistance R offered by a conductor depends on the following factors :

(*i*) It varies directly as its length, l.

(*ii*) It varies inversely as the cross-section A of the conductor.

(*iii*) It depends on the nature of the material.

(*iv*) It also depends on the temperature of the conductor.



Smaller l
Larger A
Low R

Larger l
Smaller A
Greater R

Fig. 1.3.

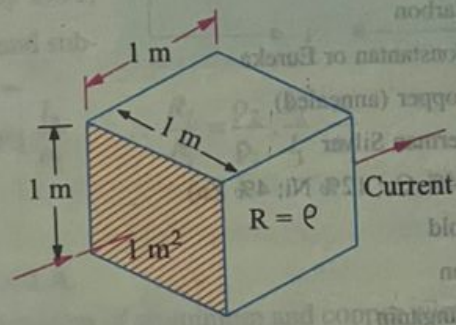


Fig. 1.4

Neglecting the last factor for the time being, we can say that

$$R \propto \frac{l}{A} \quad \text{or} \quad R = \rho \frac{l}{A} \qquad \text{...(i)}$$

where ρ is a constant depending on the nature of the material of the conductor and is known as its ***specific resistance* or *resistivity***.

If in Eq. (*i*), we put

$$l = 1 \text{ metre} \quad \text{and} \quad A = 1 \text{ metre}^2, \text{ then } R = \rho \quad \text{(Fig. 1.4)}$$

Hence, specific resistance of a material may be defined as ***the resistance between the opposite faces of a metre cube of that material.***

1.7. Units of Resistivity

From Eq. (*i*), we have $\rho = \dfrac{AR}{l}$

In the S.I. system of units,

$$\rho = \frac{A \text{ metre}^2 \times R \text{ ohm}}{l \text{ metre}} = \frac{AR}{l} \text{ ohm-metre}$$

Hence, the unit of resistivity is ohm-metre (Ω-m).

It may, however, be noted that resistivity is sometimes expressed as so many ohm per m^3. Although, it is incorrect to say so but it means the same thing as ohm-metre.

If l is in centimetres and A in cm^2, then ρ is in ohm-centimetre (Ω-cm).

Values of resistivity and temperature coefficients for various materials are given in Table 1.2. The resistivities of commercial materials may differ by several per cent due to impurities etc.

Table 1.2. Resistivities and Temperature Coefficients

Material	*Resistivity in ohm-metre at 20°C ($\times 10^{-8}$)*	*Temperature coefficient at 20°C ($\times 10^{-4}$)*
Aluminium, commercial	2.8	40.3
Brass	6 – 8	20
Carbon	3000 – 7000	–5
Constantan or Eureka	49	+0.1 to –0.4
Copper (annealed)	1.72	39.3
German Silver (84% Cu; 12% Ni; 4% Zn)	20.2	2.7
Gold	2.44	36.5
Iron	9.8	65
Manganin (84% Cu ; 12% Mn ; 4% Ni)	44 – 48	0.15
Mercury	95.8	8.9
Nichrome (60% Cu ; 25% Fe ; 15% Cr)	108.5	1.5
Nickel	7.8	54
Platinum	9 – 15.5	36.7
Silver	1.64	38
Tungsten	5.5	47
Amber	5×10^{14}	
Bakelite	10^{10}	
Glass	$10^{10} - 10^{12}$	
Mica	10^{15}	
Rubber	10^{16}	
Shellac	10^{14}	
Sulphur	10^{15}	

Example 1.3. *A coil consists of 2000 turns of copper wire having a cross-sectional area of 0.8 mm². The mean length per turn is 80 cm and the resistivity of copper is 0.02 μΩ–m. Find the resistance of the coil and power absorbed by the coil when connected across 110 V d.c. supply.*

(F.Y. Engg. Pune Univ. May 1990)

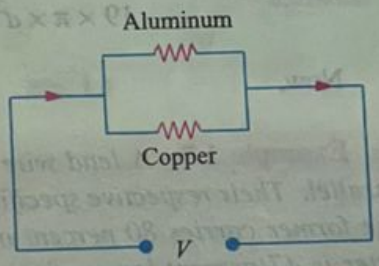
Solution. Length of the coil, $l = 0.8 \times 2000 = 1600$ m ; $A = 0.8 \text{ mm}^2 = 0.8 \times 10^{-6} \text{ m}^2$.

$$R = \rho \frac{l}{A} = 0.02 \times 10^{-6} \times 1600/0.8 \times 10^{-6} = \mathbf{40\ \Omega}$$

$$\text{Power absorbed} = V^2 / R = 110^2/40 = \mathbf{302.5\ W}$$

Example 1.4. *An aluminium wire 7.5 m long is connected in a parallel with a copper wire 6 m long. When a current of 5 A is passed through the combination, it is found that the current in the aluminium wire is 3 A. The diameter of the aluminium wire is 1 mm. Determine the diameter of the copper wire. Resistivity of copper is 0.017 μΩ-m ; that of the aluminium is 0.028 μΩ-m.*

(F.Y. Engg. Pune Univ. May 1991)



Solution. Let the subscript 1 represent aluminium and subscript 2 represent copper.

$$R_1 = \rho \frac{l_1}{a_1} \text{ and } R_2 = \rho_2 \frac{l_2}{a_2} \qquad \therefore \frac{R_2}{R_1} = \frac{\rho_2}{\rho_1} \cdot \frac{l_2}{l_1} \cdot \frac{a_1}{a_2}$$

$$\therefore \quad a_2 = a_1 \cdot \frac{R_1}{R_2} \cdot \frac{\rho_2}{\rho_1} \cdot \frac{l_2}{l_1} \qquad \ldots(i)$$

Now $\quad I_1 = 3 \text{ A} ; I_2 = 5 - 3 = 2 \text{ A}.$

If V is the common voltage across the parallel combination of aluminium and copper wires, then

$$V = I_1 R_1 = I_2 R_2 \quad \therefore \quad R_1/R_2 = I_2/I_1 = 2/3$$

$$a_1 = \frac{\pi d^2}{4} = \frac{\pi \times 1^2}{4} = \frac{\pi}{4} \text{ mm}^2$$

Substituting the given values in Eq. (*i*), we get

$$a_2 = \frac{\pi}{4} \times \frac{2}{3} \times \frac{0.017}{0.028} \times \frac{6}{7.5} = 0.2544 \text{ m}^2$$

$$\therefore \quad \pi \times d_2^2/4 = 0.2544 \quad \text{or} \quad d_2 = \mathbf{0.569\ mm}$$

Example 1.5. (*a*) *A rectangular carbon block has dimensions 1.0 cm × 1.0 cm × 50 cm. (i) What is the resistance measured between the two square ends ? (ii) between two opposing rectangular faces / Resistivity of carbon at 20°C is 3.5 × 10⁻⁵ Ω-m.*

(*b*) *A current of 5 A exists in a 10-Ω resistance for 4 minutes (i) how many coulombs and (ii) how many electrons pass through any section of the resistor in this time ? Charge of the electron = 1.6 × 10⁻¹⁹ C.*

(M.S. Univ. Baroda)

Solution.

(*a*) (*i*) $\quad R = \rho\, l/A$

Here, $\quad A = 1 \times 1 = 1 \text{ cm}^2 = 10^{-4} \text{ m}^2 ; l = 0.5 \text{ m}$

$\therefore \quad R = 3.5 \times 10^{-5} \times 0.5/10^{-4} = \mathbf{0.175\ \Omega}$

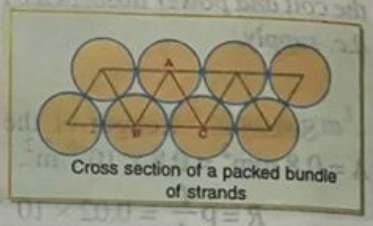
(*ii*) Here, $\quad l = 1 \text{ cm}; A = 1 \times 50 = 50 \text{ cm}^2 = 5 \times 10^{-3} \text{ m}^2$

$R = 3.5 \times 10^{-5} \times 10^{-2}/5 \times 10^{-3} = \mathbf{7 \times 10^{-5}\ \Omega}$

(*b*) (*i*) $Q = It = 5 \times (4 \times 60) = 1200$ C

(*ii*) $n = \dfrac{Q}{e} = \dfrac{1200}{1.6 \times 10^{-19}} = (75 \times 10^{20})$ electrons

Example 1.6. *Calculate the resistance of 1 km long cable composed of 19 stands of similar copper conductors, each strand being 1.32 mm in diameter. Allow 5% increase in length for the 'lay' (twist) of each strand in completed cable. Resistivity of copper may be taken as 1.72×10^{-8} Ω-m.*



Cross section of a packed bundle of strands

Solution. Allowing for twist, the length of the strands.

$$= 1000 \text{ m} + 5\% \text{ of } 1000 \text{ m} = 1050 \text{ m}$$

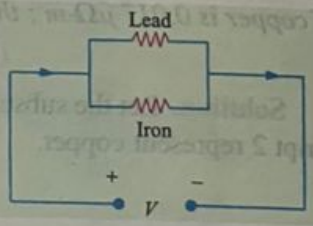
Area of cross-section of 19 strands of copper conductors is

$$19 \times \pi \times d^2/4 = 19\,\pi \times (1.32 \times 10^{-3})^2/4 \text{ m}^2$$

Now,
$$R = \rho \frac{l}{A} = \frac{1.72 \times 10^{-8} \times 1050 \times 4}{19\pi \times 1.32^2 \times 10^{-6}} = 0.694\ \Omega$$

Example 1.7. *A lead wire and an iron wire are connected in parallel. Their respective specific resistances are in the ratio 49 : 24. The former carries 80 percent more current than the latter and the latter is 47 percent longer than the former. Determine the ratio of their cross sectional areas.*

(Elect. Engg. Nagpur Univ. 1993)



Solution. Let suffix 1 represent lead and suffix 2 represent iron. We are given that

$$\rho_1/\rho_2 = 49/24; \text{ if } i_2 = 1, i_1 = 1.8; \text{ if } l_1 = 1, l_2 = 1.47$$

Now,
$$R_1 = \rho_1 \frac{l_1}{A_1} \quad \text{and} \quad R_2 = \rho_2 \frac{l_2}{A_2}$$

Since the two wires are in parallel, $i_1 = V/R_1$ and $i_2 = V/R_2$

$$\therefore \quad \frac{i_2}{i_1} = \frac{R_1}{R_2} = \frac{\rho_1 l_1}{A_1} \times \frac{A_2}{\rho_2 l_2}$$

$$\therefore \quad \frac{A_2}{A_1} = \frac{i_2}{i_1} \times \frac{\rho_2 l_2}{\rho_1 l_1} = \frac{1}{1.8} \times \frac{24}{49} \times 1.47 = 0.4$$

Example 1.8. *A piece of silver wire has a resistance of 1 Ω. What will be the resistance of manganin wire of one-third the length and one-third the diameter, if the specific resistance of manganin is 30 times that of silver.* **(Electrical Engineering-I, Delhi Univ.)**

Solution. For silver wire, $R_1 = \dfrac{l_1}{A_1}$; For manganin wire, $R = \rho_2 \dfrac{l_2}{A_2}$

$$\therefore \quad \frac{R_2}{R_1} = \frac{\rho_2}{\rho_1} \times \frac{l_2}{l_1} \times \frac{A_1}{A_2}$$

Now $\quad A_1 = \pi d_1^2/4 \quad$ and $\quad A_2 = \pi\, d_2^2/4 \qquad \therefore \quad A_1/A_2 = d_1^2/d_2^2$

$$\therefore \quad \frac{R_2}{R_1} = \frac{\rho_2}{\rho_1} \times \frac{l_2}{l_1} \times \left(\frac{d_1}{d_2}\right)^2$$

$$R_1 = 1\ \Omega;\ l_2/l_1 = 1/3,\ (d_1/d_2)^2 = (3/1)^2 = 9;\ \rho_2/\rho_1 = 30$$

$$\therefore \quad R_2 = 1 \times 30 \times (1/3) \times 9 = 90\ \Omega$$